Random Variables and Probability Distributions

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Discrete and continuous data; Empirical vs theoretical probability, probability rules.
- **Leads to:** Expected value and variance of a random variable; Probability density function (PDF); Probability mass function (PMF); Bernoulli and binomial distributions; Normal distribution; Log-normal distribution; Uniform distribution; Uncertainty in robotics; Belief.

1. Overview

Key point: A random variable turns the outcomes of a random experiment into numbers; a **probability distribution** (G-1571) says how likely each number is, and it comes as a PMF, a PDF or a CDF.

The family of probability distribution functions

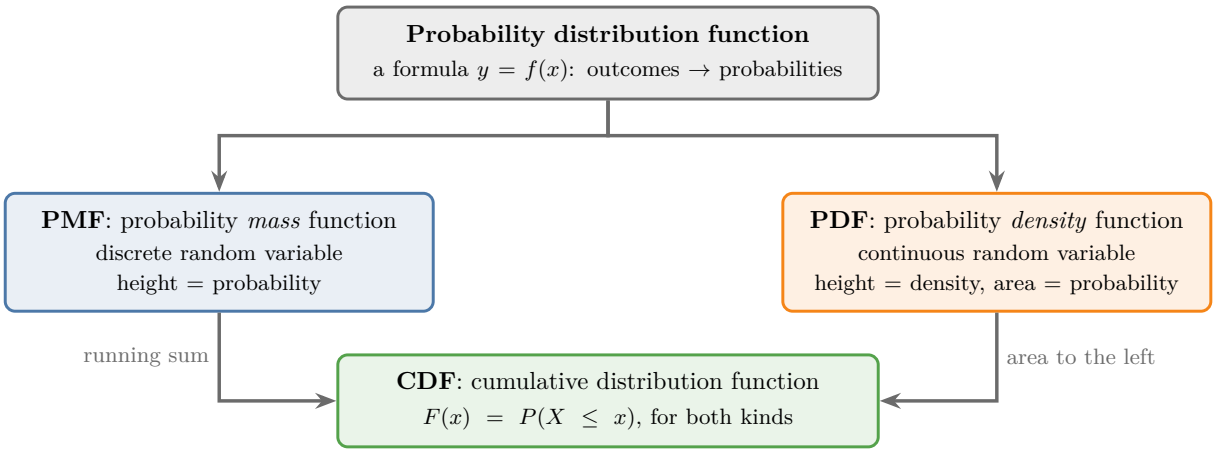


Figure 1: The family of probability distribution functions

Figure 1 is the map of this Note and the three after it. A **probability distribution function** (G-1570) is a formula that links every outcome to its probability. The function has two kinds, one for each kind of **random variable** (G-1620), and both lead to a third function, the CDF.

This Note builds the base:

- random variables (section 2);
- distributions as tables (section 3);

- why we want a formula instead (sections 4 and 5);
- the famous distributions and why they matter (sections 6 and 7);
- their **parameters** (G-1449) (section 8).

The three functions themselves are taught in the **probability mass function** (G-1572), the **probability density function** (G-1568) and the **cumulative distribution function** (G-515), for a discrete variable and for a continuous variable.

2. Random variables

> **Key point:** An algebra variable holds one unknown value; a **random variable** (G-1620) can take several values, and which one appears is decided by chance.

2.1 Algebra variables and random variables

> **Key point:** We solve for an algebra variable; we cannot solve for a random variable, only describe how likely each of its values is.

In algebra, a variable stands for one unknown value. In $x + 5 = 10$ we simplify and get $x = 5$: there is exactly one answer.

Statistics and probability use a different kind of variable. A **random variable** (G-1620) is the set of possible numerical values of a **random experiment** (G-1610). A random variable differs from an algebra variable in two ways:

- **A random variable holds many values.** A die can show any of 1 to 6.
- **The value that appears is random.** We cannot solve for it; we can only say how likely each value is.

2.2 Random experiments

> **Key point:** A random experiment has an outcome we cannot predict; the **random variable** (G-1620) writes its outcomes as numbers.

A **random experiment** (G-1610) is any experiment whose outcome is random: tossing a coin, rolling a die, picking a student and asking their CGPA. Figure 2 turns three such experiments into random variables.

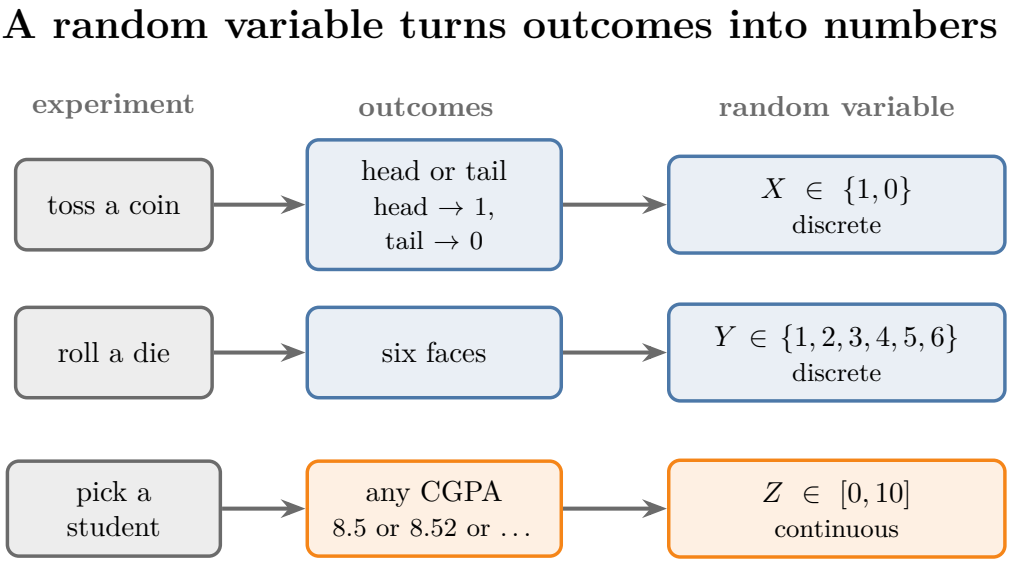


Figure 2: Three random experiments and their random variables

- **Coin toss:** writing head as 1 and tail as 0 gives $X = \{1, 0\}$ (curly braces list the values of a set), assuming a fair (unbiased) coin.
- **Die roll:** $Y = \{1, 2, 3, 4, 5, 6\}$.
- **CGPA of a random student:** Z can be any number from 0 to 10.

By convention, a random variable gets a **capital letter** (X, Y) and an algebra variable a small one (x). A particular value of X is written with the small letter: the statement $X = x$ reads as "the random variable X takes the value x". The set of all possible outcomes is the **sample space** (G-1729; see an example with two dice).

Turning outcomes into numbers pays off in the notation. Instead of the sentence "the probability that the two dice add up to 7", we write $P(X = 7)$; "the probability that the sum is at most 4" becomes $P(X \leq 4)$. Every question about the experiment becomes a short statement about X.

> **Extra:** Strictly, a random variable is a **function**: it assigns a number to each outcome of the sample space (MML §6.1.2). Head → 1 and tail → 0 is such a function. The numbers are our choice: head → 100 and tail → 703 would also be a valid random variable, only a less convenient one. "The set of possible values" describes what the function can output. Thinking of it as a function explains why the same experiment can have several random variables: rolling two dice can give "the sum", "the larger face" or "the number of sixes".

2.3 Discrete and continuous random variables

> **Key point:** A discrete random variable takes separate values (a die never shows 1.5); a continuous one can take any value in a range (a CGPA can be 8.527).

Random variables (G-1620) come in the same two kinds as numerical data (see discrete

and continuous data):

- A **discrete random variable** (G-617) takes separate values. The coin gives 0 or 1, the die 1 to 6; a value of 1.5 or 1.6 can never appear.
- A **continuous random variable** (G-466) can take any value in a range. A CGPA can be 8.5, 8.52, 8.527 or 6.532: every decimal between 0 and 10 is possible.

The difference decides which function describes the variable (Figure 1): a PMF for a discrete variable, a PDF for a continuous one.

3. Probability distributions as tables

> **Key point:** A probability distribution lists every possible outcome of a **random variable** (G-1620) with its probability, like a frequency table with probabilities in place of counts.

A **probability distribution** (G-1571) is a list of all the possible outcomes of a random variable, each with its probability. The list looks like the **frequency distribution table** (G-807), with probabilities instead of counts.

Coin toss. Two outcomes, equally likely:

x	1 (head)	0 (tail)
$P(X = x)$	1/2	1/2

One die. Six outcomes, all **equiprobable** (G-704), meaning equally likely:

x	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

3.1 The sum of two dice

> **Key point:** Two dice give 36 equally likely pairs; counting the pairs for each sum gives probabilities from 1/36 (sums 2 and 12) up to 6/36 (sum 7).

When we roll two dice and add the faces, the sum can be anything from 2 (a 1 and a 1) to 12 (a 6 and a 6). The 36 pairs of faces are equally likely; an example with two dice (Figure 2 there) draws them as a 6 by 6 grid with the sum in each cell. Here the sums are not equally likely, because some sums can be made in more ways than others. Figure 3 counts them. Watch the orange cells of the grid: each sum lights up a diagonal, and the diagonal through 7 is the longest.

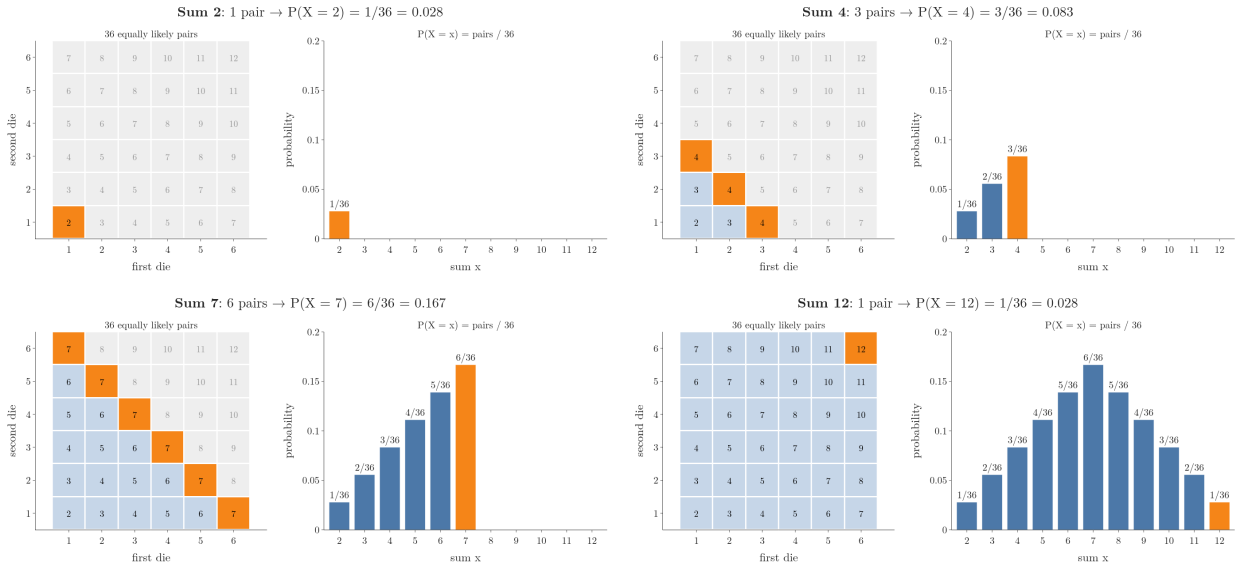


Figure 3: The 36 equally likely pairs of two dice. For each sum, the pairs that give it light up (orange) and its bar, pairs / 36, is added on the right.

1. **In words:** the probability of a sum is the number of pairs that give it, divided by the 36 pairs.
2. **Formula:**
$$P(X = x) = \frac{\text{number of pairs with sum } x}{36}$$
3. **Example:** the sum 7 comes from (1, 6), (2, 5), (3, 4), (4, 3), (5, 2) and (6, 1), six pairs:
$$P(X = 7) = \frac{6}{36} \approx 0.167$$

   The sum 2 comes only from (1, 1):
$$P(X = 2) = \frac{1}{36} \approx 0.028$$

Counting every sum gives the whole distribution:

Sum x	Pairs	$P(X = x)$
2	1	1/36
3	2	2/36
4	3	3/36
5	4	4/36
6	5	5/36
7	6	6/36
8	5	5/36
9	4	4/36
10	3	3/36

Sum x	Pairs	$P(X = x)$
11	2	2/36
12	1	1/36

The most likely sum is 7; the least likely are 2 and 12. The counts rise by one up to 7 and fall by one after it.

4. From a table to a function

> **Key point:** A table fails when there are too many outcomes or infinitely many; a formula $y = f(x)$ from outcome to probability works in both cases, and it can be graphed.

A table works for a coin or two dice. A table breaks down in two situations:

- **Too many outcomes.** With 10 dice the sum runs from 10 to 60: 51 sums. Each die has 6 faces, so the number of equally likely combinations is:

$$6^{10} = 60{,}466{,}176$$

  Writing that table by hand is tedious (Figure 4).

- **A continuous random variable.** A CGPA can be any number between 0 and 10, so the outcomes cannot even be listed.

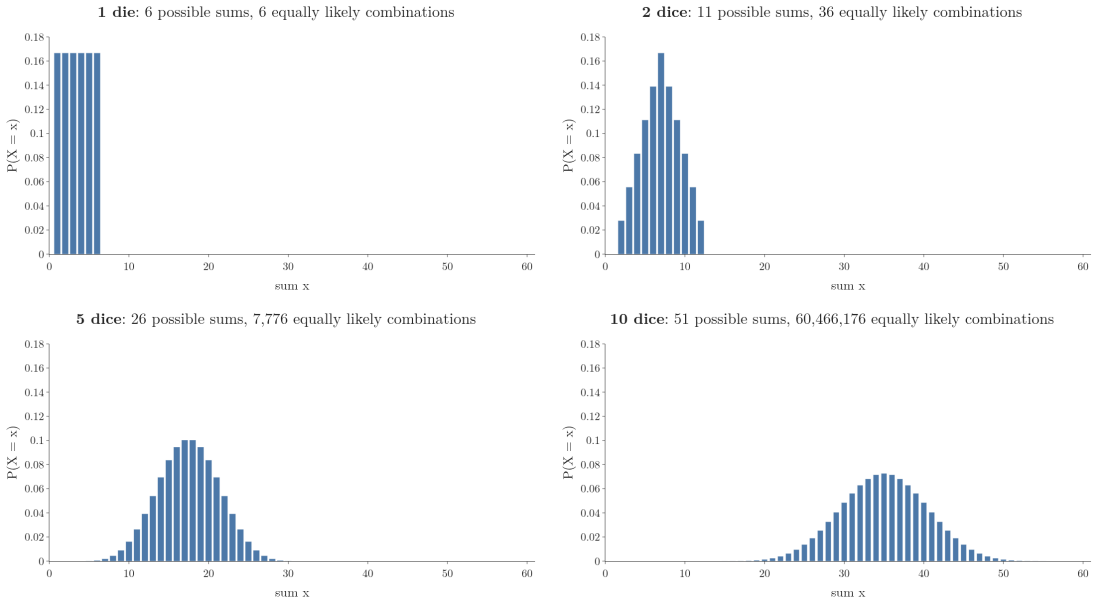


Figure 4: The exact distribution of the sum of 1, 2, 3, 5 and 10 dice. The number of sums and of combinations grows fast; the graph still shows the shape at a glance.

Figure 4 also shows the way out. The table for 10 dice would have 51 rows, but its graph is one simple hump, and a simple shape can be described by a formula. The way out is a function. Call the outcome x and its probability y. Instead of a table, we look for a formula $y = f(x)$ that **models** the relationship between them. A formula has two benefits:

1. **Any outcome at once:** we plug in any x and the formula returns its y.
2. **A graph:** a formula can be plotted, and the graph shows the shape of the distribution at a glance.

The formula $y = f(x)$ is the **probability distribution function** (G-1570): a mathematical function that links every possible outcome of a random variable to its probability. For one die:

$$f(x) = 1/6 \quad \text{for } x = 1, \ldots, 6$$

$$f(x) = 0 \quad \text{for any other } x$$

> **Extra:** "**Probability distribution** (G-1571)" (the table, or the idea) and "probability distribution function" (the formula) are used interchangeably in most books. Both mean the description of how likely each value is; the table and the formula are two ways of writing it down.

5. Discrete and continuous graphs

> **Key point:** The graph of a discrete distribution has values only at separate points; the graph of a continuous one is an unbroken curve.

The two kinds of **random variable** (G-1620) give two kinds of graph (Figure 5):

- **Discrete:** a die has values only at 1, 2, ..., 6. At 1.5 there is nothing, because 1.5 is not in the sample space. The graph is a set of separate bars or sticks with gaps between them.
- **Continuous:** a CGPA has a value at every point from 0 to 10, so the graph is an unbroken curve. We can pick any x and read off its y.

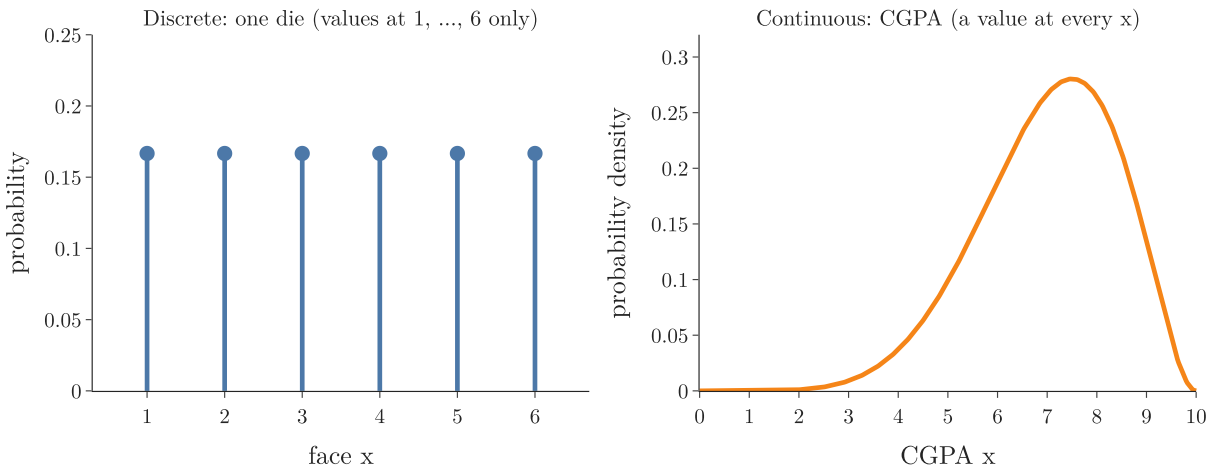


Figure 5: The distribution of one die (separate values) and of CGPA (an unbroken curve)

The unbroken curve is what a histogram turns into. Measure many students and stack their CGPAs in bins, as in histograms for a numerical feature; with more students and narrower bins, the tops of the bars settle onto a smooth curve. Bars whose area is probability (Figure 1 there) animates this step. The histogram and the curve are both distributions of the same feature, and the curve has three advantages:

- **No empty bins.** A bin in which we happened to measure nobody does not mean such a student is impossible. The curve still gives that range a small probability.
- **Any range.** The curve gives the probability of a CGPA between 7.021 and 7.317 as easily as between 7 and 8; a histogram can answer only for whole bins.
- **Little data is enough.** A curve needs only a few numbers, such as a mean (the average) and a standard deviation (how far values typically lie from the mean; see standard deviation), which are the **parameters** (G-1449) of section 8, and those can be estimated from a small sample.

The two graphs also differ in what the height means. For the die, the height is a probability. For the CGPA curve the y axis is labelled **probability density** (G-1569), which is not a probability; what the density at a point means explains why.

5.1 PMF, PDF and CDF

> **Key point:** For a discrete variable the function is a PMF (mass); for a continuous one it is a PDF (density); either can be turned into a CDF (cumulative).

The **probability distribution function** (G-1570) is the umbrella term for three functions (Figure 1):

- **Probability mass function (PMF)** (G-1572): the probability distribution function of a **discrete random variable** (G-1620), such as the die.

- **Probability density function (PDF)** (G-1568): the probability distribution function of a **continuous** random variable, such as the CGPA.
- **Cumulative distribution function (CDF)** (G-515): built from either of them; it gives the probability that the variable is **at most** x.

"PDF" is ambiguous: both "probability **distribution** function" and "probability **density** function" shorten to it. In these Notes, **PDF always means the density function**; the distribution function is always written in full.

6. Famous distributions

> **Key point:** Data from very different fields often follows one of a few common shapes; these famous distributions have names, known formulas and known **parameters** (G-1449).

When people in many fields (science, medicine, engineering, economics) plotted the distributions of their data, the same few shapes kept appearing. These are the **famous probability distributions** (G-753). Figure 6 shows eight of them: discrete ones as bars, continuous ones as curves.

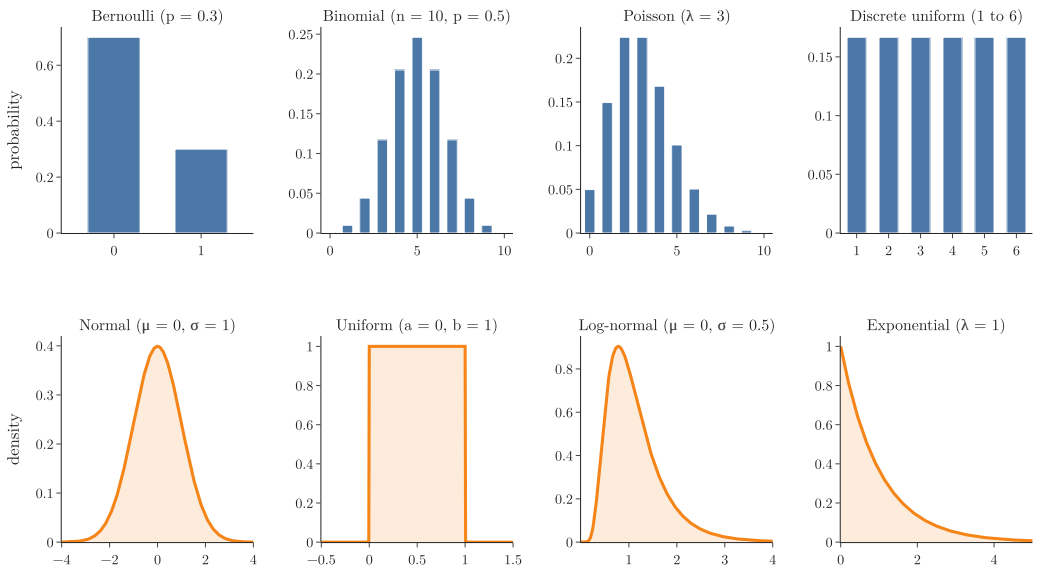


Figure 6: Eight famous distributions: four discrete (top) and four continuous (bottom), with their parameters

- **Discrete:** Bernoulli, binomial, Poisson, discrete uniform.
- **Continuous:** normal, uniform, log-normal, exponential.
- **Others** we will meet later: beta, chi-square, Pareto.

When we plot the distribution of a real **feature** (G-772; one variable of the data, one column of the table), there is a good chance it resembles one of these. The **normal distribution** (G-1343) is the most famous; the **68-95-99.7 rule** (G-53) was already used to find outliers.

The binomial distribution appeared in why voting works. Each named distribution gets its own Note later.

7. Why distributions matter

> **Key point:** A distribution shows the shape of the data, and if that shape matches a famous distribution, everything already known about it applies to our data.

The graph of a distribution is only x (outcomes) against y (their probabilities), yet it gives us two things.

1. The shape of the data. The graph shows where the values lie, which are common and which are rare:

- If most marks in a class (out of 10) sit around 5 to 6, the class has many weak students.
- If most marks sit around 8, with few below 7, most of the class is strong.
- If a country's salary distribution has a big mass of high earners, it is probably a developed country; if most of the mass sits at low salaries, poverty is likely widespread.

2. Ready-made knowledge. If our data's shape matches a famous distribution, we can apply everything known about that distribution to our data. The **normal distribution** (G-1343) has been studied for a long time and its mathematics is fully worked out. Knowing that a feature is normal, we immediately know, for example, that about 95% of its values lie within two standard deviations of the mean.

8. Parameters of a distribution

> **Key point:** Parameters are the tuning knobs of a distribution: numbers such as μ and σ that set its location, scale and shape.

Every famous distribution comes with a few numbers written next to its name in Figure 6: μ and σ, a and b, n and p, λ. These are its **parameters** (G-1449): numerical values that determine the shape, location and scale of the distribution. Changing them changes the graph, like turning a knob on a radio.

Figure 7 turns the two knobs of the **normal distribution** (G-1343):

- **μ, the location (mean):** moves the curve left or right without changing its shape.
- **σ, the scale (standard deviation):** widens or narrows the curve. A wider curve is lower, because the total area under it stays the same (see area under the curve is probability).

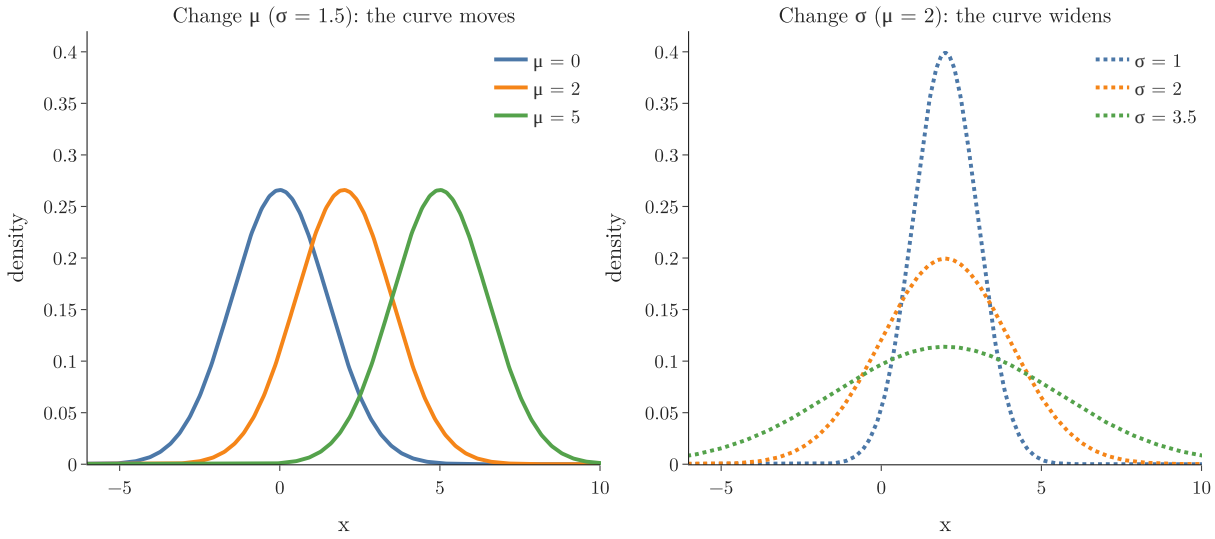


Figure 7: Changing the mean moves the normal curve; changing the standard deviation widens it

Different distributions have different sets of parameters. Whenever we study a distribution, we study what each of its parameters does.

The word "parameter" is the same as in parameters and statistics, where a parameter is a number that describes the population. The two meanings meet: if a population is normal, its mean μ and standard deviation σ are exactly the knobs of its normal curve. Estimating those knobs from a sample is the idea behind **density estimation** (G-586).

9. Summary

Idea	Meaning	Example	Why it matters
Random variable	The numerical outcomes of a random experiment	die: $Y = \{1, \ldots, 6\}$	a question becomes a short statement such as $P(X = 7)$
Discrete random variable	Takes separate values	coin, die, sum of two dice	described by a PMF
Continuous random variable	Takes any value in a range	CGPA from 0 to 10	described by a PDF
Probability distribution	Every outcome with its probability	two dice: $P(X = 7) = 6/36$	shows which values are common and which are rare
Probability distribution function	A formula $y = f(x)$ from outcome to probability	$f(x) = 1/6$ for $x = 1, \ldots, 6$	gives any outcome at once and can be graphed
PMF / PDF / CDF	Discrete / continuous / cumulative	die / CGPA / $P(X \leq x)$	the kind of variable decides which function to use
Parameters	Knobs that set location, scale and shape	normal: μ, σ	a few numbers, estimated from a sample, describe the whole curve

- **Random variables** (G-1620) get capital letters; their values get small letters, so $X = x$ reads "the random variable X takes the value x".
- A table cannot list too many outcomes or a continuous range; a function can, and it can be plotted.
- In these Notes, PDF means **probability density function** (G-1568), because "probability distribution function" shortens to PDF too.
- A distribution shows the shape of the data; matching a famous one lets us reuse its known mathematics (for a normal feature, about 95% of values lie within two standard deviations of the mean).

10. Sources

Built from

- CampusX, "Session 40 - Probability Distribution Functions - PDF, PMF & CDF | DSMP 2023", YouTube, https://www.youtube.com/watch?v=C_QAURbgBqY
- Khan Academy, "Random variables", YouTube, https://www.youtube.com/watch?v=3v9w79NhsfI
- StatQuest with Josh Starmer, "The Main Ideas behind Probability Distributions", YouTube, https://www.youtube.com/watch?v=oI3hZJqXJuc

Other references

- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 6.1.2 (random variable as a function).

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Probability distribution (G-1571)	A list (or curve) of every possible outcome of a random variable with its probability, adding up to 1; it says how likely each outcome is, so the chance of any event can be worked out from it.
Probability distribution function (G-1570)	A formula $y = f(x)$ giving the probability of each outcome; the umbrella term for PMF and PDF.
Random variable (G-1620)	The possible numerical outcomes of a random experiment; strictly, a function from outcomes to numbers.
Parameters (of a distribution) (G-1449)	The numbers, such as μ and σ, that set a distribution's location, scale and shape.
Discrete random variable (G-617)	A random variable that takes separate values, such as a die's face.
Continuous random variable (G-466)	A random variable that can take any value in a range, such as a CGPA.
Equiprobable (G-704)	Equally likely: every outcome has the same probability, as the six faces of a fair die, so each has probability 1 divided by the number of outcomes.
Famous probability distributions (G-753)	Common named shapes such as normal, uniform, binomial and Poisson.
Probability mass function (PMF) (G-1572)	The function that gives the probability of each exact value of a discrete random variable, such as $P(X = 3) = 1/6$ for a fair die; its probabilities add up to 1.
Probability density function (PDF) (G-1568)	A curve showing how likely each value is; areas under it are probabilities.
Cumulative distribution function (CDF) (G-515)	For every value x, the probability that the random variable comes out at most x: $P(X \leq x)$.
Random experiment (G-1610)	An experiment whose outcome cannot be predicted, such as a coin toss.

Term	Meaning
Sample space (G-1729)	The set of all possible outcomes of an experiment, such as $\{1, \ldots, 6\}$ for a die; every event is a subset of it, so probabilities are worked out inside it.
Frequency distribution table (G-807)	A table of each value or category with the number of times it occurs.
Probability density (G-1569)	The height of a continuous distribution's curve; compares how likely nearby values are.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Normal distribution (G-1343)	A symmetric, bell-shaped distribution: most values sit near the mean and they get rarer the further out we go; its mean and standard deviation set its centre and width.
68-95-99.7 rule (empirical rule) (G-53)	In a column shaped like a bell curve (a normal distribution), about 68.3%, 95.4% and 99.7% of values lie within 1, 2 and 3 standard deviations of the mean.
Density estimation (G-586)	Working out, from observed data, how likely each value of a random variable is: estimating its probability density function (PDF).